1) 25% Utilizando **la identidad de Abel** encuentre una segunda solución linealmente independiente de la ecuación diferencial $(2x+1)^2 y'' + 14(2x+1)y' + 36y = 0$, sabiendo que $y_1 = \dfrac{\ln(2x+1)}{(2x+1)^3}$ es una solución de dicha ecuación.

Solución

$(2x+1)^2 y'' + 14(2x+1)y' + 36y = 0$, $y_1 = \dfrac{\ln(2x+1)}{(2x+1)^3}$

$y'' + \dfrac{14}{(2x+1)} y' + \dfrac{36}{(2x+1)^2} y = 0$ 05

$y_2 = y_1 \int \dfrac{e^{-\int P(x)dx}}{(y_1)^2} dx$

$P(x) = \dfrac{14}{(2x+1)}$ 05

$e^{-\int P(x)dx} = e^{-\int \frac{14}{(2x+1)}dx} = e^{-7\int \frac{2}{(2x+1)}dx} = e^{-7\ln(2x+1)} = \dfrac{1}{(2x+1)^7}$

$y_2 = y_1 \int \dfrac{e^{-\int P(x)dx}}{(y_1)^2} dx \rightarrow y_2 = \dfrac{\ln(2x+1)}{(2x+1)^3} \int \dfrac{\frac{1}{(2x+1)^7}}{\frac{\ln^2(2x+1)}{(2x+1)^6}} dx$

$y_2 = \dfrac{\ln(2x+1)}{(2x+1)^3} \int \dfrac{1}{(2x+1)\ln^2(2x+1)} dx$ 05

$u = \ln(2x+1)$

$du = \dfrac{2}{2x+1}dx \rightarrow \dfrac{du}{2} = \dfrac{1}{2x+1}dx$

$y_2 = \dfrac{\ln(2x+1)}{(2x+1)^3} \int \dfrac{1}{u^2}\left(\dfrac{du}{2}\right)$

$y_2 = \left(\dfrac{1}{2}\right)\dfrac{\ln(2x+1)}{(2x+1)^3} \int u^{-2} du \quad \rightarrow \quad y_2 = \left(\dfrac{1}{2}\right)\dfrac{\ln(2x+1)}{(2x+1)^3}\left(\dfrac{u^{-1}}{-1}\right)$

$y_2 = -\left(\dfrac{1}{2}\right)\dfrac{\ln(2x+1)}{(2x+1)^3}\left(\dfrac{1}{u}\right) \rightarrow \quad y_2 = -\left(\dfrac{1}{2}\right)\dfrac{\ln(2x+1)}{(2x+1)^3}\left(\dfrac{1}{\ln(2x+1)}\right)$

$y_2 = -\left(\dfrac{1}{2}\right)\dfrac{1}{(2x+1)^3}$ 10

Una segunda solución es $y = \dfrac{1}{(2x+1)^3}$

2) 15% Resolver la siguiente ecuación diferencial . $y^{(4)}+2y''+2y=0$

$$y^{(4)}+2y''+2y=0 \Rightarrow m^4+2m^2+2=0 \Rightarrow m^4+2+2m^2=0$$

$$m^4+2\sqrt{2}m^2+2+2m^2-2\sqrt{2}m^2=0$$

$$\downarrow \qquad \downarrow$$

$$m^2 \qquad \sqrt{2}$$

$$\left(m^2+\sqrt{2}\right)^2-\left(2\sqrt{2}-2\right)m^2=0$$

$$\left[\left(m^2+\sqrt{2}\right)+\sqrt{2\sqrt{2}-2}\,m\right]\left[\left(m^2+\sqrt{2}\right)-\sqrt{2\sqrt{2}-2}\,m\right]=0$$

05

$$m^2+\sqrt{2\sqrt{2}-2}\,m+\sqrt{2}=0 \qquad m^2-\sqrt{2\sqrt{2}-2}\,m+\sqrt{2}=0$$

$$m^2+\sqrt{2\sqrt{2}-2}\,m+\sqrt{2}=0$$

$$m=-\frac{\sqrt{2\sqrt{2}-2}}{2}\pm\sqrt{\frac{2\sqrt{2}-2}{4}-\sqrt{2}}$$

$$m=-\frac{\sqrt{2\sqrt{2}-2}}{2}\pm\sqrt{\frac{2\sqrt{2}-2-4\sqrt{2}}{4}}$$

$$m=-\frac{\sqrt{2\sqrt{2}-2}}{2}\pm\frac{\sqrt{-2\sqrt{2}-2}}{2}$$

$$m=-\frac{\sqrt{2\sqrt{2}-2}}{2}\pm\frac{\sqrt{2\sqrt{2}+2}}{2}i$$

$$\alpha=-\frac{\sqrt{2\sqrt{2}-2}}{2}\ ,\quad \beta=\frac{\sqrt{2\sqrt{2}+2}}{2}$$

03

$$m^2-\sqrt{2\sqrt{2}-2}\,m+\sqrt{2}=0$$

$$m=\frac{\sqrt{2\sqrt{2}-2}}{2}\pm\sqrt{\frac{2\sqrt{2}-2}{4}-\sqrt{2}}$$

$$m=\frac{\sqrt{2\sqrt{2}-2}}{2}\pm\sqrt{\frac{2\sqrt{2}-2-4\sqrt{2}}{4}}$$

$$m=\frac{\sqrt{2\sqrt{2}-2}}{2}\pm\frac{\sqrt{-2\sqrt{2}-2}}{2}$$

$$m=\frac{\sqrt{2\sqrt{2}-2}}{2}\pm\frac{\sqrt{2\sqrt{2}+2}}{2}i$$

$$\alpha=\frac{\sqrt{2\sqrt{2}-2}}{2}\ ,\quad \beta=\frac{\sqrt{2\sqrt{2}+2}}{2}$$

03

$$y=e^{-\frac{\sqrt{2\sqrt{2}-2}}{2}x}\left[c_1\cos\left(\frac{\sqrt{2\sqrt{2}+2}}{2}x\right)+c_2 sen\left(\frac{\sqrt{2\sqrt{2}+2}}{2}x\right)\right]+e^{\frac{\sqrt{2\sqrt{2}-2}}{2}x}\left[c_3\cos\left(\frac{\sqrt{2\sqrt{2}+2}}{2}x\right)+c_4 sen\left(\frac{\sqrt{2\sqrt{2}+2}}{2}x\right)\right]$$

04

3) 30% Utilizando el método del anulador resuelva la siguiente ecuación diferencial

$$2y''' - 3y'' + 2y' - 3y = sen^2\left(\frac{x}{2}\right) - x$$

Una solución

$$2y''' - 3y'' + 2y' - 3y = sen^2\left(\frac{x}{2}\right) - x$$

$$2m^3 - 3m^2 + 2m - 3 = 0$$

$$(2m-3)(m^2+1) = 0$$

$$m = \frac{3}{2} \quad ; m = \pm i$$

$$\boxed{y = C_1 e^{\frac{3}{2}x} + C_2\cos(x) + C_3 sen(x)} \qquad 05$$

$$f(x) = sen^2\left(\frac{x}{2}\right) - x$$

$$f(x) = \frac{1}{2} - \frac{1}{2}\cos(x) - x$$

$$\boxed{anulador\ \ de\ \ f(x) \quad D^2(D^2+1)} \qquad 05$$

Para y_p

$m^2(m^2+1) = 0$

$m = 0 \ , \ m = 0$

$m = \pm i$

Para y_c

$2m^3 - 3m^2 + 2m - 3 = 0$

$m = \frac{3}{2} \quad ; m = \pm i$

$$\boxed{y_p = A + Bx + Cx\cos(x) + Dxsen(x)} \qquad 05$$

$$\boxed{y_P' = \quad B \ + C\cos(x) - Cxsen(x) + Dsen(x) + Dx\cos(x)}$$

$$y_P'' = \quad -Csen(x) - Csen(x) - Cx\cos(x) + D\cos(x) + D\cos(x) - Dxsen(x)$$

$$\boxed{y_P'' = \quad -2Csen(x) - Cx\cos(x) + 2D\cos(x) - Dxsen(x)}$$

$$y_P''' = \quad -2C\cos(x) - C\cos(x) + Cxsen(x) - 2Dsen(x) - Dsen(x) - Dx\cos(x)$$

$$\boxed{y_P''' = \quad -3C\cos(x) + Cxsen(x) - 3Dsen(x) - Dx\cos(x)}$$

$$2y_p''' = \quad -6C\cos(x) + 2Cxsen(x) - 6Dsen(x) - 2Dx\cos(x)$$

$$-3y_p'' = \quad -6D\cos(x) + 3Dxsen(x) + 6Csen(x) + 3Cx\cos(x)$$

$$2y_p' = \quad 2B \ + \ 2C\cos(x) - 2Cxsen(x) + 2Dsen(x) + 2Dx\cos(x)$$

$$-3y_p = -3A - 3Bx \qquad -3Dxsen(x) \qquad -3Cx\cos(x)$$

$$\frac{1}{2} - \frac{1}{2}\cos(x) - x = (2B - 3A) - 3Bx + (-4C - 6D)\cos(x) + (-4D + 6C)sen(x)$$

$$-3B=-1\rightarrow \boxed{B=\frac{1}{3}}$$

$$2B-3A=\frac{1}{2}\rightarrow 2\left(\frac{1}{3}\right)-3A=\frac{1}{2}\rightarrow 4-18A=3\rightarrow \boxed{A=\frac{1}{18}}$$

$$-4C-6D=-\frac{1}{2}\rightarrow \boxed{8C+12D=1}$$

$$\boxed{-4D+6C=0}$$

$$8C+12D=1$$

$$\underline{18C-12D=0}$$

$$26C=1\rightarrow \boxed{C=\frac{1}{26}}$$

$$-4D+6C=0\rightarrow D=\frac{3}{2}C\rightarrow D=\frac{3}{2}\left(\frac{1}{26}\right)$$

$$\boxed{D=\frac{3}{52}}$$

$A=\frac{1}{18}$	$B=\frac{1}{3}$	$C=\frac{1}{26}$	$D=\frac{3}{52}$

05

$$\boxed{y_p=\frac{1}{18}+\frac{1}{3}x+\frac{1}{26}x\cos(x)+\frac{3}{52}xsen(x)}$$ 05

$$y=y_c+y_P$$

$$\boxed{y=C_1e^{\frac{3}{2}x}+C_2\cos(x)+C_3sen(x)+\frac{1}{18}+\frac{1}{3}x+\frac{1}{26}x\cos(x)+\frac{3}{52}xsen(x)}$$ 05

4) 30% Resolver la ecuación diferencial

$$xy'' - y' + \left(\frac{1}{x}\right)y = 2x\ln\left(\sqrt{x}\right),\ x > 0\ \ ;\ \ y_{(1)} = 2\ \ ,\ \ y_{\left(e^2\right)} = 8e^2$$

Una solución

1°) *Cálculo de* $y_c : x^2y'' - xy' + y = 0$

$$y = x^m\ ,\ \ y' = mx^{m-1}\ \ ,\ \ \ y''' = \left(m^2 - m\right)x^{m-2}$$

$$x^2\left[\left(m^2 - m\right)x^{m-2}\right] - x\left[mx^{m-1}\right] + x^m = 0$$

$$x^m\left[m^2 - m - m + 1\right] = 0 \Rightarrow m^2 - 2m + 1 = 0 \Rightarrow \left(m-1\right)^2 = 0$$

$$m_1 = 1,\ \ m_2 = 1$$

$$y_1 = x\ \ ,\ \ \ y_2 = x\ln(x)$$

$y_c = c_1x + c_2x\ln(x)$ 05

2°) *Cálculo de* $y_p : y'' - \left(\frac{1}{x}\right)y' + \left(\frac{1}{x^2}\right)y = \ln(x)$

$y_p = u_{1(x)}y_1 + u_{2(x)}y_2 \Rightarrow y_p = u_{1(x)}x + u_{2(x)}x\ln(x)\ ,\ \ f(x) = \ln(x)$ 05

$$W=\begin{vmatrix} x & x\ln(x) \\ 1 & \ln(x)+1 \end{vmatrix}=x\ln(x)+x-x\ln(x)\Rightarrow \boxed{W=x}$$

$$W_1=\begin{vmatrix} 0 & x\ln(x) \\ \ln(x) & \ln(x)+1 \end{vmatrix}\Rightarrow \boxed{W_1=-x\left[\ln(x)\right]^2}$$

$$W_2=\begin{vmatrix} x & 0 \\ 1 & \ln(x) \end{vmatrix}\Rightarrow \boxed{W_2=x\ln(x)}$$

$$u_{1(x)}=\int\frac{W_1}{W}dx\Rightarrow u_{1(x)}=\int\frac{-x\left[\ln(x)\right]^2}{x}dx$$

$$u_{1(x)}=-\int\left[\ln(x)\right]^2dx=-\left[x\left[\ln(x)\right]^2-\int x\left[2\ln(x)\right]\frac{dx}{x}\right]=-\left[x\left[\ln(x)\right]^2-2\int\ln(x)dx\right]$$

$$\boxed{\begin{aligned} u&=\left[\ln(x)\right]^2 \\ d&=2\ln(x)\frac{dx}{x} \\ dv&=dx \\ v&=\int dx \\ v&=x \end{aligned}}\qquad \boxed{\begin{aligned} u&=\ln(x) \\ du&=\frac{dx}{x} \\ dv&=dx \\ v&=x \end{aligned}}$$

$$=-\left[x\left[\ln(x)\right]^2-2\left(x\ln(x)-\int x\left(\frac{dx}{x}\right)\right)\right]$$

$$=-\left[x\left[\ln(x)\right]^2-2\left(x\ln(x)-\int dx\right)\right]$$

$$=-\left[x\left[\ln(x)\right]^2-2\left(x\ln(x)-x\right)\right]$$

$$u_{1(x)}=-x\left[\ln(x)\right]^2+2x\ln(x)-x$$

$$u_{2(x)}=\int\frac{W_2}{w}dx=\int\frac{x\ln(x)}{x}dx=\int\ln(x)dx$$

$$u_{2(x)}=x\ln(x)-x$$

$$y_p=u_{1(x)}y_1+u_{2(x)}y_2$$

$$y_p=\left[-x\ln^2(x)+2x\ln(x)-2x\right]x+\left[x\ln(x)-x\right]x\ln(x)$$

$$y_p=-x^2\ln^2(x)+2x^2\ln(x)-2x^2+x^2\ln^2(x)-x^2\ln(x)$$

$$y_p=x^2\ln(x)-2x^2$$

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3°) *Cálculo de* $y = y_c + y_p$

$y_{(x)} = c_1 x + c_2 x \ln(x) + [\ln(x) - 2]x^2$

03

$_{(1)} = 2$

$c_1(1) + c_2(1)\ln(1) + [\ln(1) - 2](1)^2 = 2$

$c_1 - 2 = 2 \Rightarrow \boxed{c_1 = 4}$

$y_{(x)} = c_1 x + c_2 x \ln(x) + [\ln(x) - 2]x^2$

y $y(e^2) = 8e^2 \Rightarrow 4(e^2) + c_2(e^2)\ln(e^2) + [\ln(e^2) - 2]((e^2))^2 = 8e^2$

04

$4e^2 + 2e^2 c_2 + (2-2)e^4 = 0$

$2e^2(2 + c_2) = 0 \rightarrow \boxed{c_2 = -2}$

$\boxed{y_{(x)} = 4x - 2x\ln(x) + [\ln(x) - 2]x^2}$

1) 25% Utilizando la identidad de Abel encuentre una segunda solución linealmente independiente de la ecuación diferencial $(2x+3)^2 y''+14(2x+3)y'+36y=0$, sabiendo que $y_1=\dfrac{\ln(2x+3)}{(2x+3)^3}$ es una solución de dicha ecuación.

Solución

$(2x+3)^2 y''+14(2x+3)y'+36y=0$, $y_1=\dfrac{\ln(2x+3)}{(2x+3)^3}$

$y''+\dfrac{14}{(2x+3)}y'+\dfrac{36}{(2x+3)^2}y=0$ 05

$y_2=y_1\int\dfrac{e^{-\int P(x)dx}}{(y_1)^2}dx$

$P(x)=\dfrac{14}{(2x+3)}$ 05

$e^{-\int P(x)dx}=e^{-\int\frac{14}{(2x+3)}dx}=e^{-7\int\frac{2}{(2x+3)}dx}=e^{-7\ln(2x+3)}=\dfrac{1}{(2x+3)^7}$

$y_2=y_1\int\dfrac{e^{-\int P(x)dx}}{(y_1)^2}dx\rightarrow y_2=\dfrac{\ln(2x+3)}{(2x+3)^3}\int\dfrac{\frac{1}{(2x+3)^7}}{\frac{\ln^2(2x+3)}{(2x+3)^6}}dx$

$y_2=\dfrac{\ln(2x+3)}{(2x+3)^3}\int\dfrac{1}{(2x+3)\ln^2(2x+3)}dx$ 05

$u=\ln(2x+3)$

$du=\dfrac{2}{2x+3}dx\rightarrow\dfrac{du}{2}=\dfrac{1}{2x+3}dx$

$y_2=\dfrac{\ln(2x+3)}{(2x+3)^3}\int\dfrac{1}{u^2}\left(\dfrac{du}{2}\right)$

$y_2=\left(\dfrac{1}{2}\right)\dfrac{\ln(2x+3)}{(2x+3)^3}\int u^{-2}du\quad\rightarrow\quad y_2=\left(\dfrac{1}{2}\right)\dfrac{\ln(2x+3)}{(2x+3)^3}\left(\dfrac{u^{-1}}{-1}\right)$

$y_2=-\left(\dfrac{1}{2}\right)\dfrac{\ln(2x+3)}{(2x+3)^3}\left(\dfrac{1}{u}\right)\rightarrow\quad y_2=-\left(\dfrac{1}{2}\right)\dfrac{\ln(2x+3)}{(2x+3)^3}\left(\dfrac{1}{\ln(2x+3)}\right)$

$y_2=-\left(\dfrac{1}{2}\right)\dfrac{1}{(2x+3)^3}$ 10

Una segunda solución es $y=\dfrac{1}{(2x+3)^3}$

2) 15% Resolver la siguiente ecuación diferencial . $y^{(4)}+2y''+2y=0$

Solución

$$y^{(4)}+2y''+2y=0 \Rightarrow m^4+2m^2+2=0 \Rightarrow m^4+2+2m^2=0$$

$$m^4+2\sqrt{2}m^2+2+2m^2-2\sqrt{2}m^2=0$$

$$\downarrow \qquad \downarrow$$

$$m^2 \qquad \sqrt{2}$$

$$\left(m^2+\sqrt{2}\right)^2-\left(2\sqrt{2}-2\right)m^2=0$$

$$\left[\left(m^2+\sqrt{2}\right)+\sqrt{2\sqrt{2}-2}\,m\right]\left[\left(m^2+\sqrt{2}\right)-\sqrt{2\sqrt{2}-2}\,m\right]=0$$

05

$m^2+\sqrt{2\sqrt{2}-2}\,m+\sqrt{2}=0$ $\qquad$ $m^2-\sqrt{2\sqrt{2}-2}\,m+\sqrt{2}=0$

$$m^2+\sqrt{2\sqrt{2}-2}\,m+\sqrt{2}=0$$

$$m=-\frac{\sqrt{2\sqrt{2}-2}}{2}\pm\sqrt{\frac{2\sqrt{2}-2}{4}-\sqrt{2}}$$

$$m=-\frac{\sqrt{2\sqrt{2}-2}}{2}\pm\sqrt{\frac{2\sqrt{2}-2-4\sqrt{2}}{4}}$$

$$m=-\frac{\sqrt{2\sqrt{2}-2}}{2}\pm\frac{\sqrt{-2\sqrt{2}-2}}{2}$$

$$m=-\frac{\sqrt{2\sqrt{2}-2}}{2}\pm\frac{\sqrt{2\sqrt{2}+2}}{2}i$$

$$\alpha=-\frac{\sqrt{2\sqrt{2}-2}}{2}\ ,\quad \beta=\frac{\sqrt{2\sqrt{2}+2}}{2}$$

03

$$m^2-\sqrt{2\sqrt{2}-2}\,m+\sqrt{2}=0$$

$$m=\frac{\sqrt{2\sqrt{2}-2}}{2}\pm\sqrt{\frac{2\sqrt{2}-2}{4}-\sqrt{2}}$$

$$m=\frac{\sqrt{2\sqrt{2}-2}}{2}\pm\sqrt{\frac{2\sqrt{2}-2-4\sqrt{2}}{4}}$$

$$m=\frac{\sqrt{2\sqrt{2}-2}}{2}\pm\frac{\sqrt{-2\sqrt{2}-2}}{2}$$

$$m=\frac{\sqrt{2\sqrt{2}-2}}{2}\pm\frac{\sqrt{2\sqrt{2}+2}}{2}i$$

$$\alpha=\frac{\sqrt{2\sqrt{2}-2}}{2}\ ,\quad \beta=\frac{\sqrt{2\sqrt{2}+2}}{2}$$

03

$$y=e^{-\frac{\sqrt{2\sqrt{2}-2}}{2}x}\left[c_1\cos\left(\frac{\sqrt{2\sqrt{2}+2}}{2}x\right)+c_2 sen\left(\frac{\sqrt{2\sqrt{2}+2}}{2}x\right)\right]+e^{\frac{\sqrt{2\sqrt{2}-2}}{2}x}\left[c_3\cos\left(\frac{\sqrt{2\sqrt{2}+2}}{2}x\right)+c_4 sen\left(\frac{\sqrt{2\sqrt{2}+2}}{2}x\right)\right]$$

04

3) 30% Utilizando el método del anulador resuelva la siguiente ecuación diferencial

$$5y''' - 3y'' + 5y' - 3y = \cos^2\left(\frac{x}{2}\right) + x$$

Solución

$$5y''' - 3y'' + 5y' - 3y = \cos^2\left(\frac{x}{2}\right) + x$$

$$5m^3 - 3m^2 + 5m - 3 = 0$$

$$(5m-3)(m^2+1) = 0$$

$$m = \frac{3}{5} \quad ; m = \pm i$$

$$y = C_1 e^{\frac{3}{5}x} + C_2\cos(x) + C_3 sen(x) \qquad 05$$

$$f(x) = \cos^2\left(\frac{x}{2}\right) + x$$

$$f(x) = \frac{1}{2} + \frac{1}{2}\cos(x) + x$$

$$\text{anulador de } f(x) \quad D^2(D^2+1) \qquad 05$$

Para y_p

$m^2(m^2+1) = 0$

$m = 0 \ , \ m = 0$

$m = \pm i$

Para y_c

$5m^3 - 3m^2 + 5m - 3 = 0$

$m = \frac{3}{5} \quad ; m = \pm i$

$$y_p = A + Bx + Cx\cos(x) + Dxsen(x) \qquad 05$$

$$y_P' = \quad B \quad + C\cos(x) - Cxsen(x) + Dsen(x) + Dx\cos(x)$$

$$y_P'' = \quad - Csen(x) - Csen(x) - Cx\cos(x) + D\cos(x) + D\cos(x) - Dxsen(x)$$

$$y_P'' = \quad - 2Csen(x) - Cx\cos(x) + 2D\cos(x) - Dxsen(x)$$

$$y_P''' = \quad - 2C\cos(x) - C\cos(x) + Cxsen(x) - 2Dsen(x) - Dsen(x) - Dx\cos(x)$$

$$y_P''' = \quad - 3C\cos(x) + Cxsen(x) - 3Dsen(x) - Dx\cos(x)$$

$$5y_p''' = \quad -15C\cos(x) + 5Cxsen(x) - 15Dsen(x) - 5Dx\cos(x)$$

$$-3y_p'' = \quad -6D\cos(x) + 3Dxsen(x) + 6Csen(x) + 3Cx\cos(x)$$

$$5y_p' = \quad 5B \quad + \quad 5C\cos(x) - 5Cxsen(x) + 5Dsen(x) + 5Dx\cos(x)$$

$$-3y_p = -3A - 3Bx \qquad -3Dxsen(x) \qquad -3Cx\cos(x)$$

$$\frac{1}{2} + \frac{1}{2}\cos(x) + x = (5B - 3A) - 3Bx + (-10C - 6D)\cos(x) + (-10D + 6C)sen(x)$$

$$-3B=1\rightarrow \boxed{B=-\frac{1}{3}}$$

$$5B-3A=\frac{1}{2}\rightarrow 5\left(-\frac{1}{3}\right)-3A=\frac{1}{2}\rightarrow -10-18A=3\rightarrow \boxed{A=-\frac{13}{18}}$$

$$-10C-6D=\frac{1}{2}\quad\rightarrow \boxed{-20C-12D=1}$$

$$\boxed{-10D+6C=0}\rightarrow D=\frac{3}{5}C$$

$$-20C-12\left(\frac{3}{5}C\right)=1\rightarrow -20C-\frac{36}{5}C=1\rightarrow -100C-36C=5\rightarrow \boxed{C=-\frac{5}{136}}$$

$$D=\frac{3}{5}C\rightarrow D=\frac{3}{5}\left(-\frac{5}{136}\right)=-\frac{3}{136}$$

$$\boxed{D=-\frac{3}{136}}$$

$A=-\frac{13}{18}$	$B=-\frac{1}{3}$	$C=-\frac{5}{136}$	$D=-\frac{3}{136}$

05

$$\boxed{y_p=-\frac{13}{18}-\frac{1}{3}x-\frac{5}{136}x\cos(x)-\frac{3}{136}xsen(x)}$$

05

$$y=y_c+y_P$$

$$\boxed{y=C_1e^{\frac{3}{5}x}+C_2\cos(x)+C_3sen(x)-\frac{13}{18}-\frac{1}{3}x-\frac{5}{136}x\cos(x)-\frac{3}{136}xsen(x)}$$

05

4) 30% Resolver la ecuación diferencial

$$x^3y'' - x^2y' + xy = \sqrt[3]{64}\,x^3 \ln\left(\sqrt[4]{x}\right),\ x > 0\ ;\ y_{(1)} = 2\ ,\ y_{\left(e^2\right)} = 8e^2$$

Una solución

Una solución

1°) *Cálculo de* $y_c : x^2y'' - xy' + y = 0$

$$y = x^m\ ,\ y' = mx^{m-1}\ ,\ y''' = \left(m^2 - m\right)x^{m-2}$$

$$x^2\left[\left(m^2 - m\right)x^{m-2}\right] - x\left[mx^{m-1}\right] + x^m = 0$$

$$x^m\left[m^2 - m - m + 1\right] = 0 \Rightarrow m^2 - 2m + 1 = 0 \Rightarrow (m-1)^2 = 0$$

$$m_1 = 1,\quad m_2 = 1$$

$$y_1 = x\ ,\ y_2 = x\ln(x)$$

$$\boxed{y_c = c_1x + c_2x\ln(x)}$$ 05

2°) *Cálculo de* $y_p : y'' - \left(\frac{1}{x}\right)y' + \left(\frac{1}{x^2}\right)y = \ln(x)$

$$y_p = u_{1(x)}y_1 + u_{2(x)}y_2 \Rightarrow y_p = u_{1(x)}x + u_{2(x)}x\ln(x)\ ,\ f(x) = \ln(x)$$

05

$$W=\begin{vmatrix} x & x\ln(x) \\ 1 & \ln(x)+1 \end{vmatrix}=x\ln(x)+x-x\ln(x)\Rightarrow \boxed{W=x}$$

$$W_1=\begin{vmatrix} 0 & x\ln(x) \\ \ln(x) & \ln(x)+1 \end{vmatrix}\Rightarrow \boxed{W_1=-x\left[\ln(x)\right]^2}$$

$$W_2=\begin{vmatrix} x & 0 \\ 1 & \ln(x) \end{vmatrix}\Rightarrow \boxed{W_2=x\ln(x)}$$

$$u_{1(x)}=\int\frac{W_1}{W}dx\Rightarrow u_{1(x)}=\int\frac{-x\left[\ln(x)\right]^2}{x}dx$$

$$u_{1(x)}=-\int\left[\ln(x)\right]^2dx=-\left[x\left[\ln(x)\right]^2-\int x\left[2\ln(x)\right]\frac{dx}{x}\right]=-\left[x\left[\ln(x)\right]^2-2\int\ln(x)dx\right]$$

$$\boxed{\begin{aligned} u&=\left[\ln(x)\right]^2 \\ du&=2\ln(x)\frac{dx}{x} \\ dv&=dx \\ v&=\int dx \\ v&=x \end{aligned}}$$

$$\boxed{\begin{aligned} u&=\ln(x) \\ du&=\frac{dx}{x} \\ dv&=dx \\ v&=x \end{aligned}}$$

$$=-\left[x\left[\ln(x)\right]^2-2\left(x\ln(x)-\int x\left(\frac{dx}{x}\right)\right)\right]$$

$$=-\left[x\left[\ln(x)\right]^2-2\left(x\ln(x)-\int dx\right)\right]$$

$$=-\left[x\left[\ln(x)\right]^2-2\left(x\ln(x)-x\right)\right]$$

$$u_{1(x)}=-x\left[\ln(x)\right]^2+2x\ln(x)-2x$$

$$u_{2(x)}=\int\frac{W_2}{w}dx=\int\frac{x\ln(x)}{x}dx=\int\ln(x)dx$$

$$u_{2(x)}=x\ln(x)-x$$

$$y_p=u_{1(x)}y_1+u_{2(x)}y_2$$

$$y_p=\left[-x\ln^2(x)+2x\ln(x)-2x\right]x+\left[x\ln(x)-x\right]x\ln(x)$$

$$y_p=-x^2\ln^2(x)+2x^2\ln(x)-2x^2+x^2\ln^2(x)-x^2\ln(x)$$

$$y_p=x^2\ln(x)-2x^2$$

3°) *Cálculo de* $y = y_c + y_p$

$$y_{(x)} = c_1 x + c_2 x \ln(x) + \left[\ln(x) - 2\right] x^2$$

03

$$y_{(1)} = 2$$

$$c_1 (1) + c_2 (1) \ln(1) + \left[\ln(1) - 2\right](1)^2 = 2$$

$$c_1 - 2 = 2 \Rightarrow \boxed{c_1 = 4}$$

$$y_{(x)} = c_1 x + c_2 x \ln(x) + \left[\ln(x) - 2\right] x^2$$

$$y_{(e^2)} = 8e^2 \Rightarrow 4\left(e^2\right) + c_2 \left(e^2\right) \ln\left(e^2\right) + \left[\ln\left(e^2\right) - 2\right]\left(\left(e^2\right)\right)^2 = 8e^2$$

04

$$4e^2 + 2e^2 c_2 + (2 - 2)e^4 = 0$$

$$2e^2 (2 + c_2) = 0 \rightarrow \boxed{c_2 = -2}$$

$$\boxed{y_{(x)} = 4x - 2x \ln(x) + \left[\ln(x) - 2\right] x^2}$$